LIÈGE université

Master 1 Oceanography - Coastal oceanography

Presentation of the Drifters' Results

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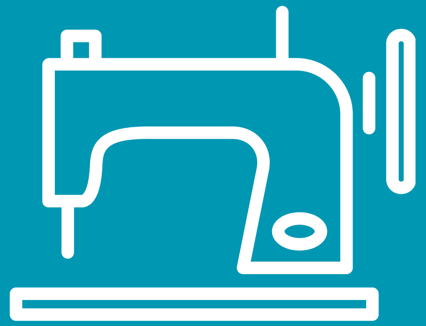
Coding



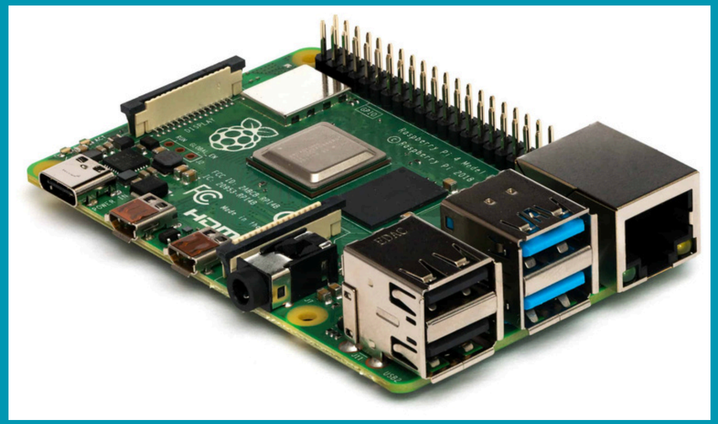
Sails

Mast

Boat

Results

Raspberry Pi



→ **Programming a microcomputer to receive the drifter's GPS coordinates**

1. Software Installation

Download Raspberry Pi Manager onto the SD card that will be used in the mini computer

2. Reduced battery consumption

Disabling the HDMI port, USB port, and Bluetooth

3. GSM Modem

Configuring the SIM card to send SMS messages via a GPS antenna

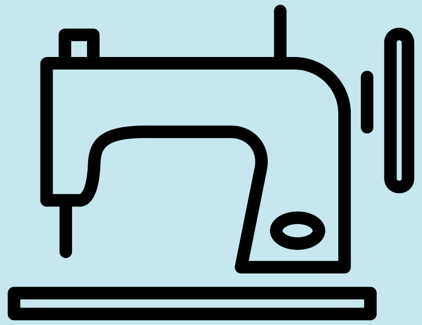
4. I-Button

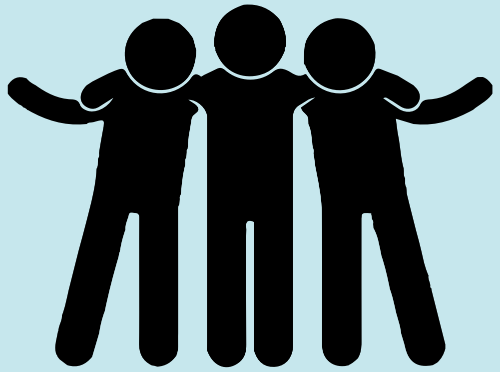
Configuring an I-Button to measure water temperature (using Windows software)

Drifter Design

→ Mast and sails built based on last year's designs

→ Skills :





→ Release of drifters in Calvi Bay

→ Records location data and temperature every minute

→ Code written in Julia to generate the graphs

Deployment of drifters

- **Deployment** on two different dates
  - May 6, 2024, between 10:27 a.m. and 3:21 p.m.
  - May 8, 2024, between 9:06 a.m. and 2:37 p.m.
- **Strategy**: All drifters are deployed in a line along a transect
- **Objective:** To observe surface currents and temperature variations between the head of the canyon and Calvi Bay.

Map showing the locations of all drifters

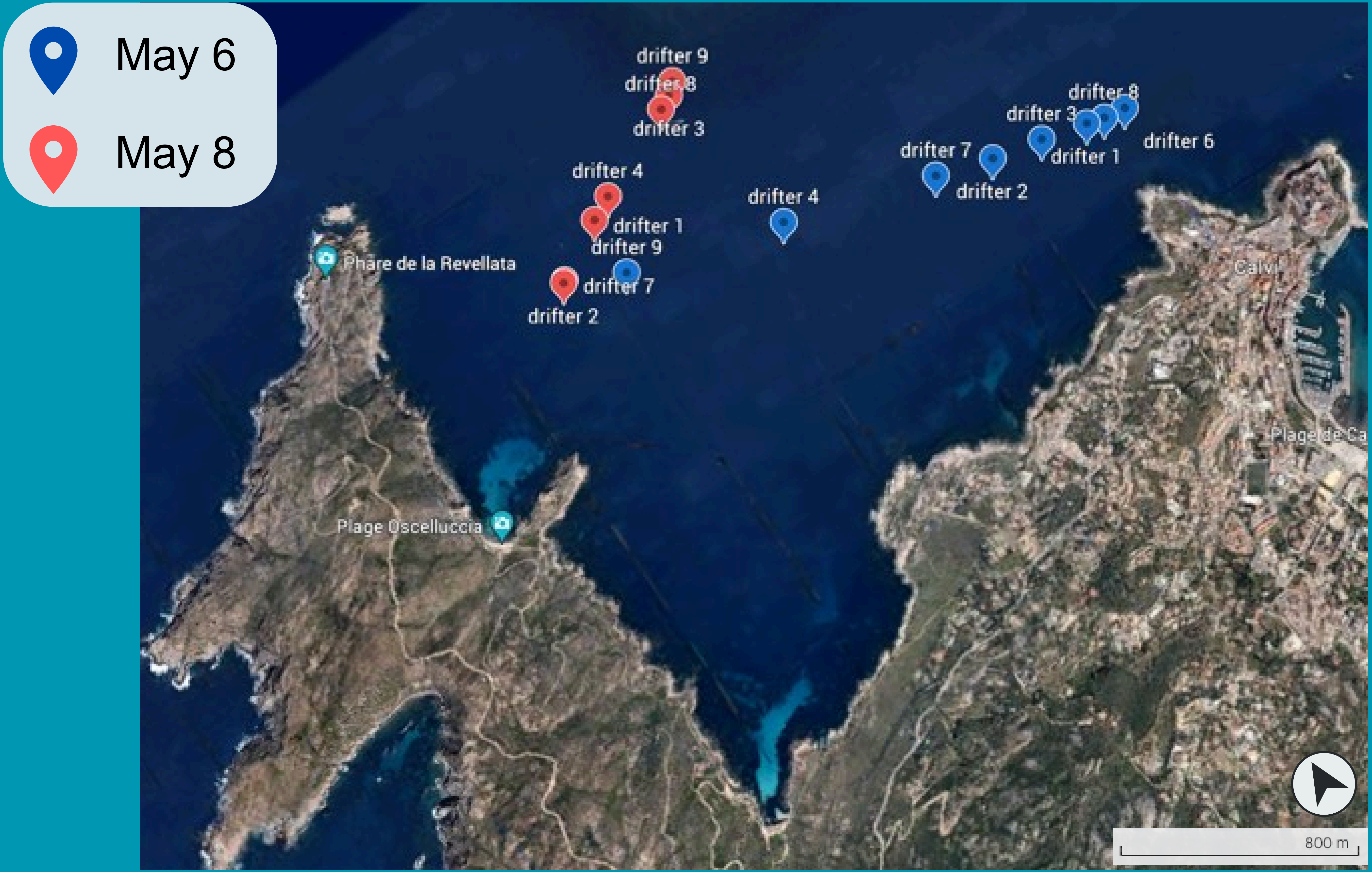


Figure 1: Map showing the release locations of the drifters

Map showing the locations of all drifters

May 6

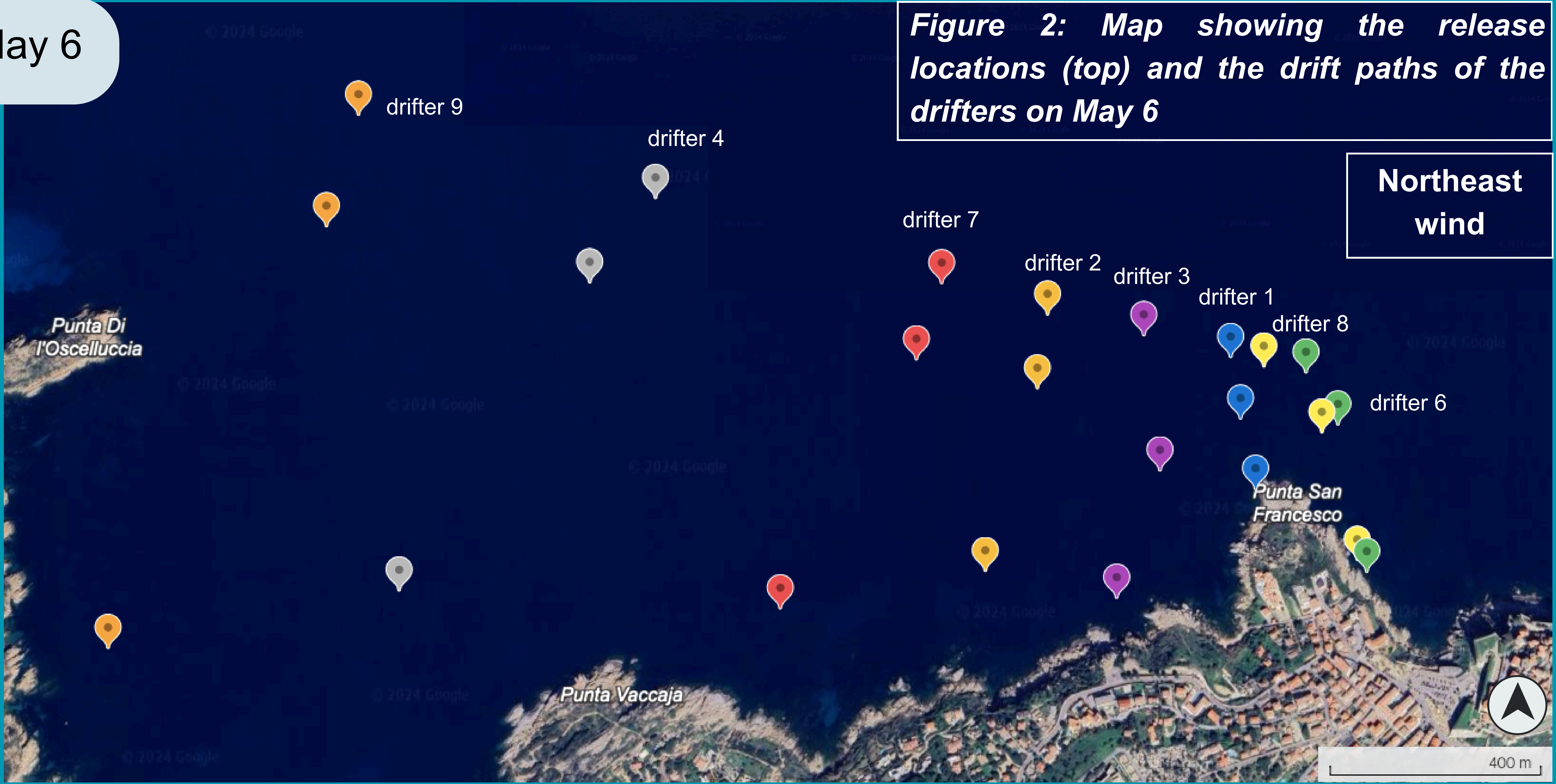


Figure 2: Map showing the release locations (top) and the drift paths of the drifters on May 6

Map showing the locations of all drifters

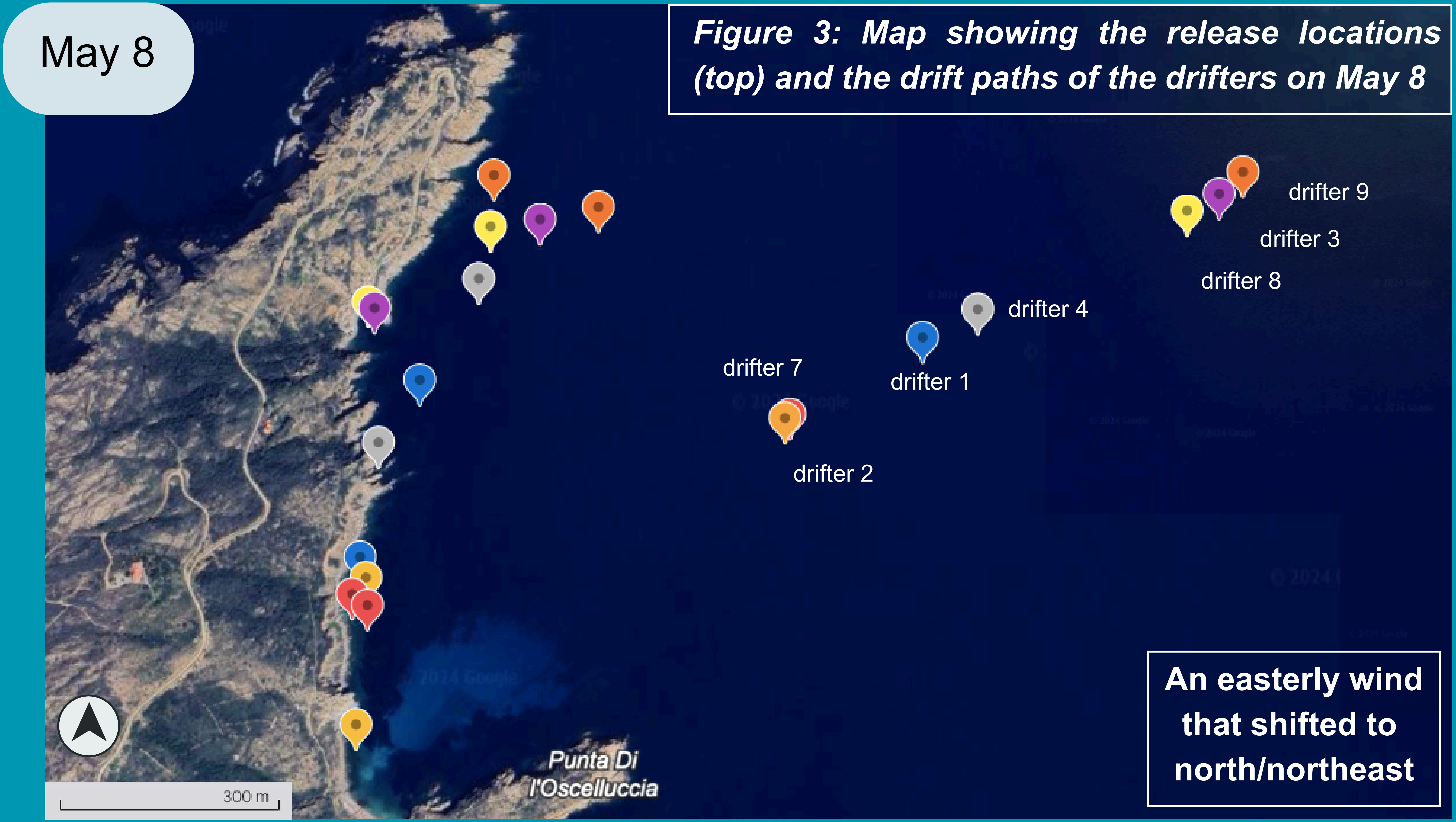


Figure 3: Map showing the release locations (top) and the drift paths of the drifters on May 8

An easterly wind that shifted to north/northeast

Analysis of Results: Drifter, May 7–6

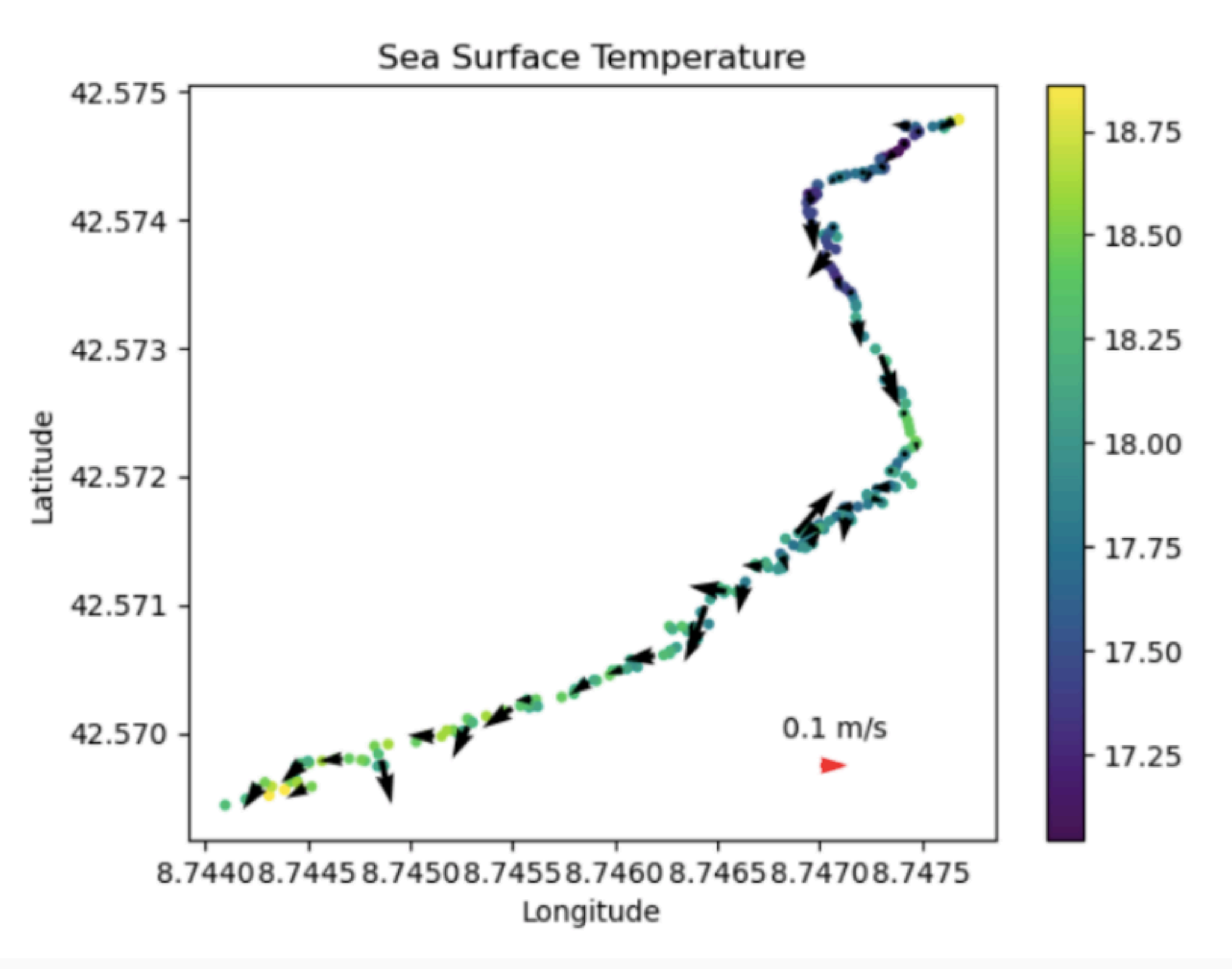


Figure 5: Surface temperatures as a function of longitude and latitude. Surface current strength is shown by black arrows.

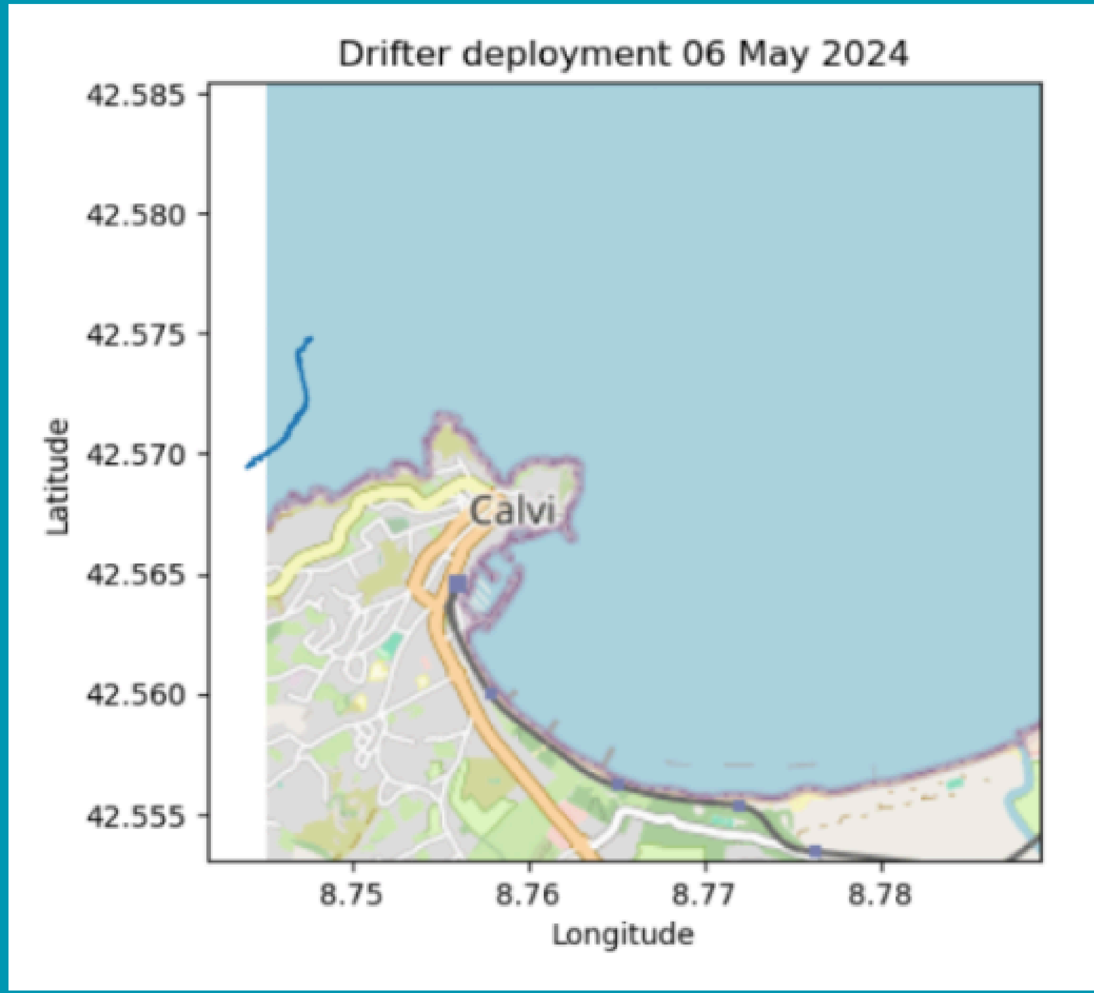


Figure 4: Map showing the drifter's movements in Calvi Bay on May 6, 2024.

- **Warmer water along the coast: +1.5°C**
- **Light currents with a maximum of 0.2 m/s**
- **Current arrows consistent with the drifter's direction**

Analysis of Results: Drifter, May 7–6

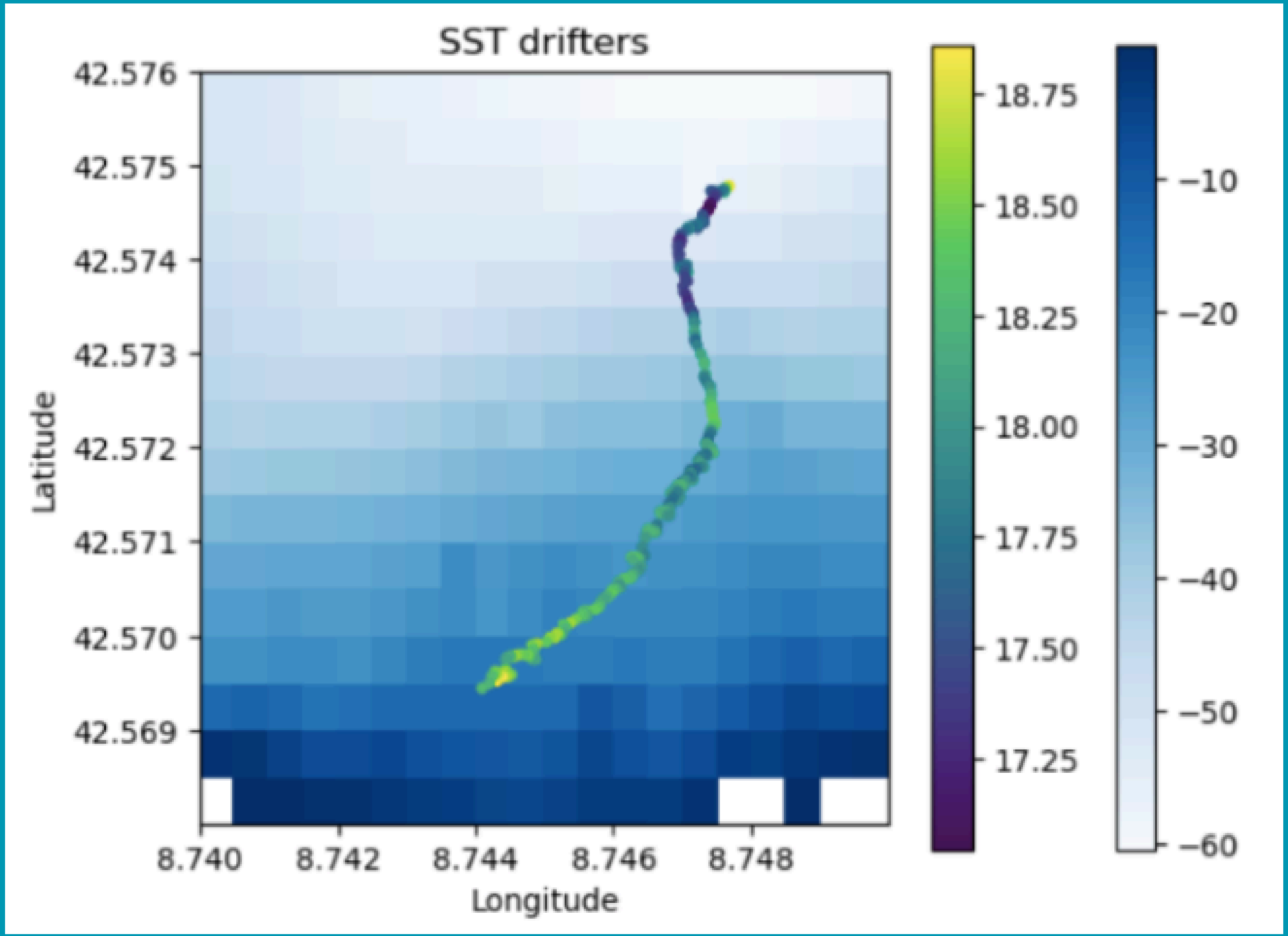


Figure 6: Surface temperatures as a function of GPS location. Bathymetry is shown using the blue scale.

- **Warmer water near the coast: +1.5°C**
-
- **Consistent with changes in bathymetry**
  - **warmer water at shallower depths**

Analysis of Results: Drifter, May 7–8

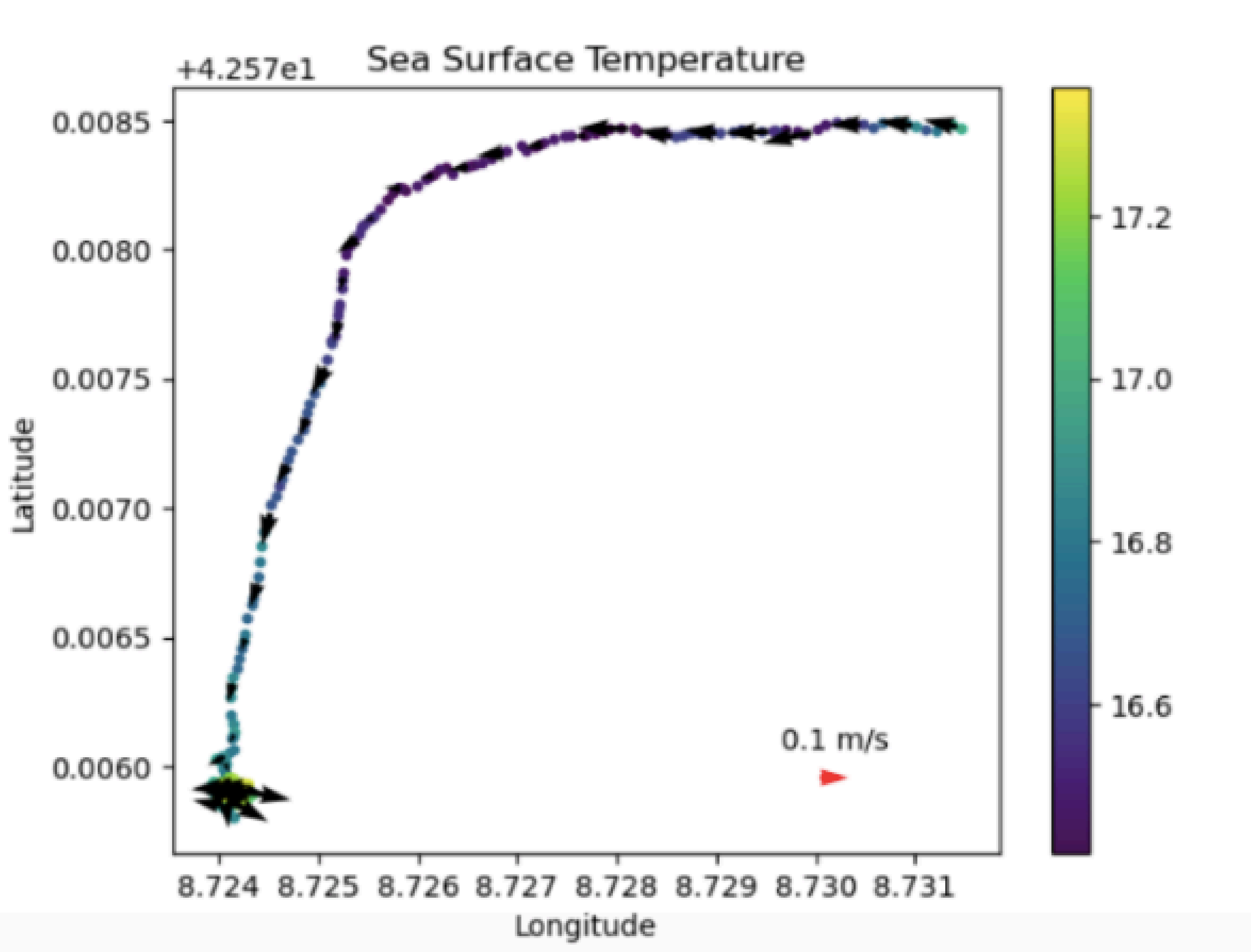


Figure 8: Surface temperatures as a function of longitude and latitude. Surface current strength is shown by black arrows.

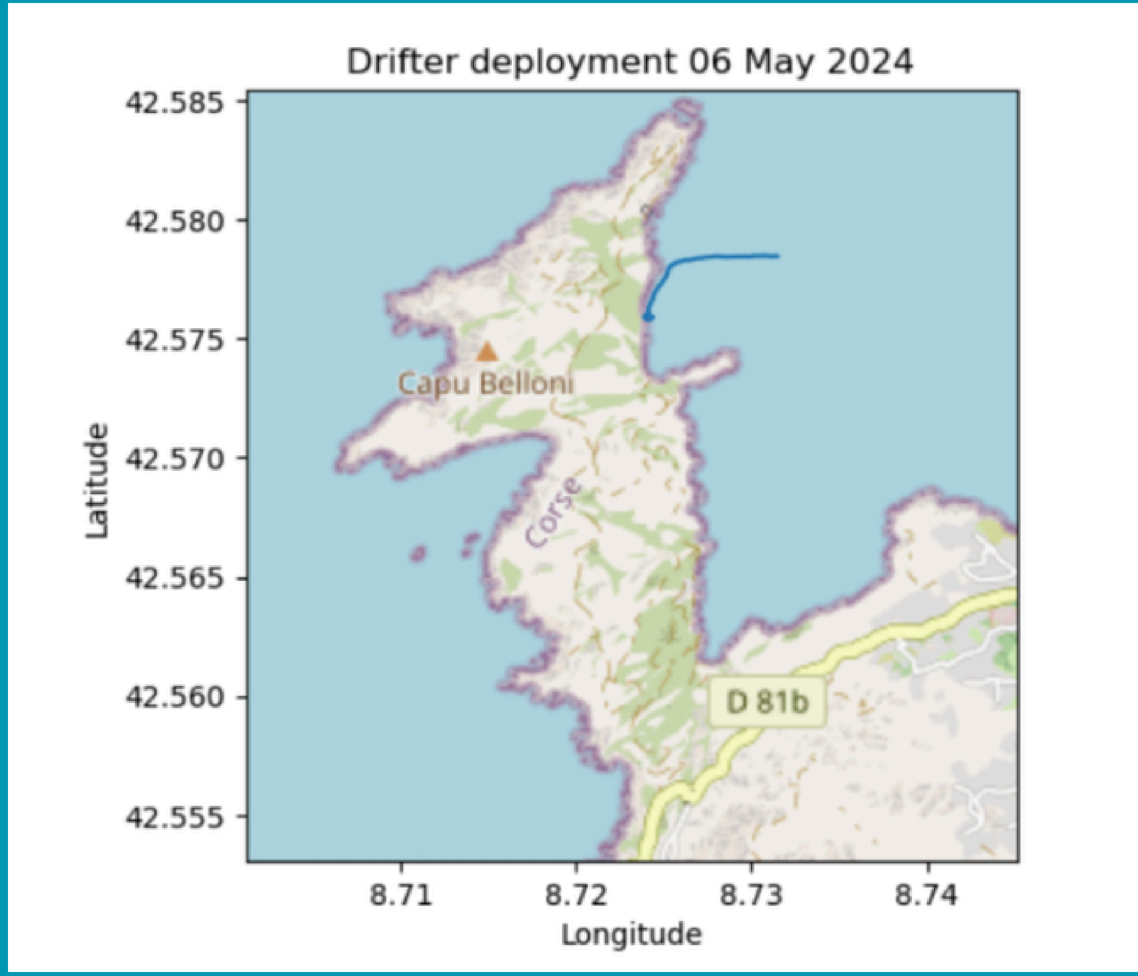


Figure 7: Map showing the drifter's movements in Calvi Bay on May 8, 2024.

- **Warmer water near the coast: +1°C**
- **Current direction consistent with the drifter's direction**
- **Very light winds**
- **Stuck at the last coordinates**

Analysis of Results: Drifter, May 7–8

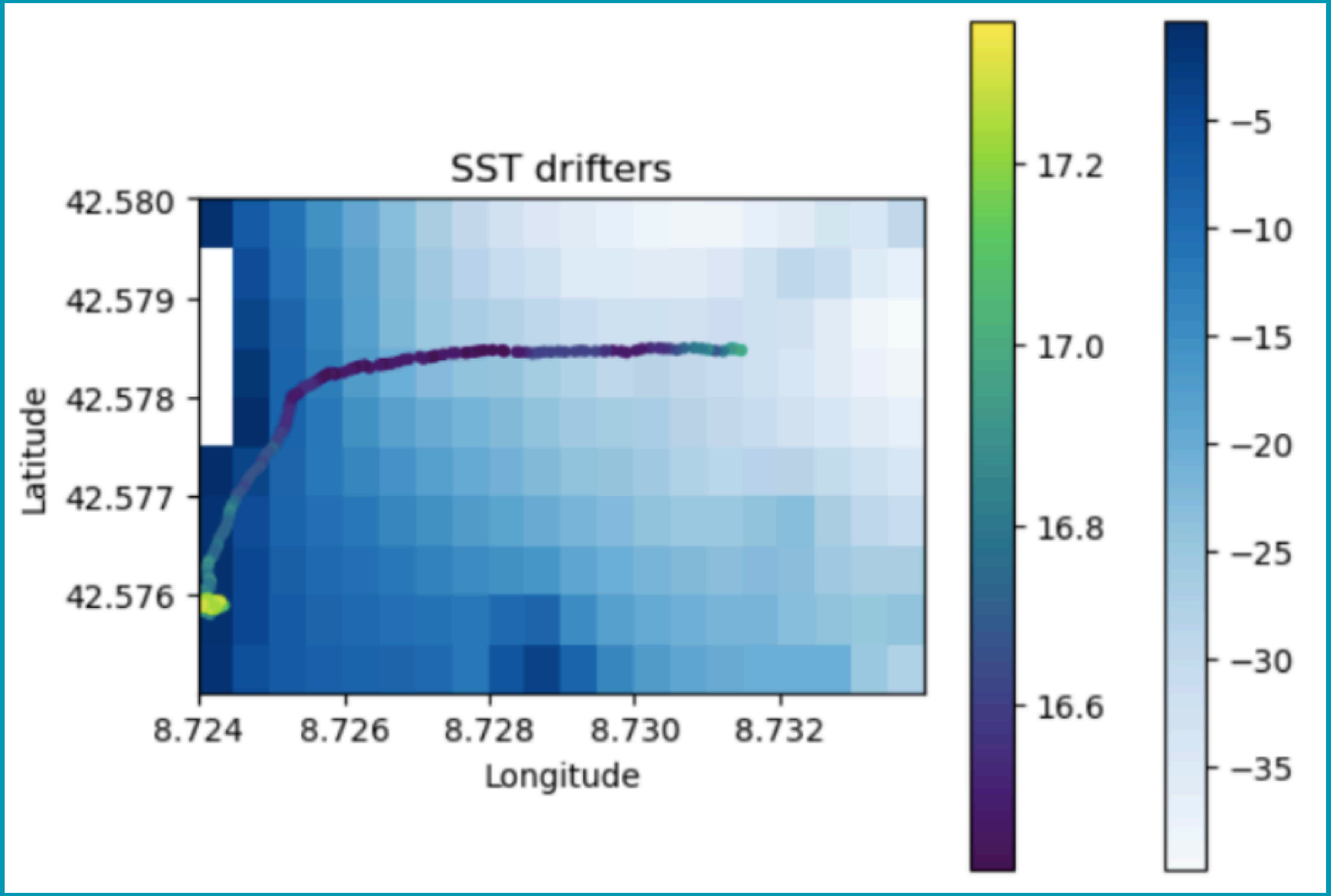


Figure 9: Surface temperatures as a function of GPS location. Bathymetry is shown using the blue scale.

- **Warmer water near the coast: +1°C**
- **Less pronounced temperature variations than on May 6**
- **Consistent with a smaller variation in bathymetry**
- **Deployment: 40 m**
- **Retrieval: 0 m --> rocks**

Analysis of Results: Drifter, May 2–6

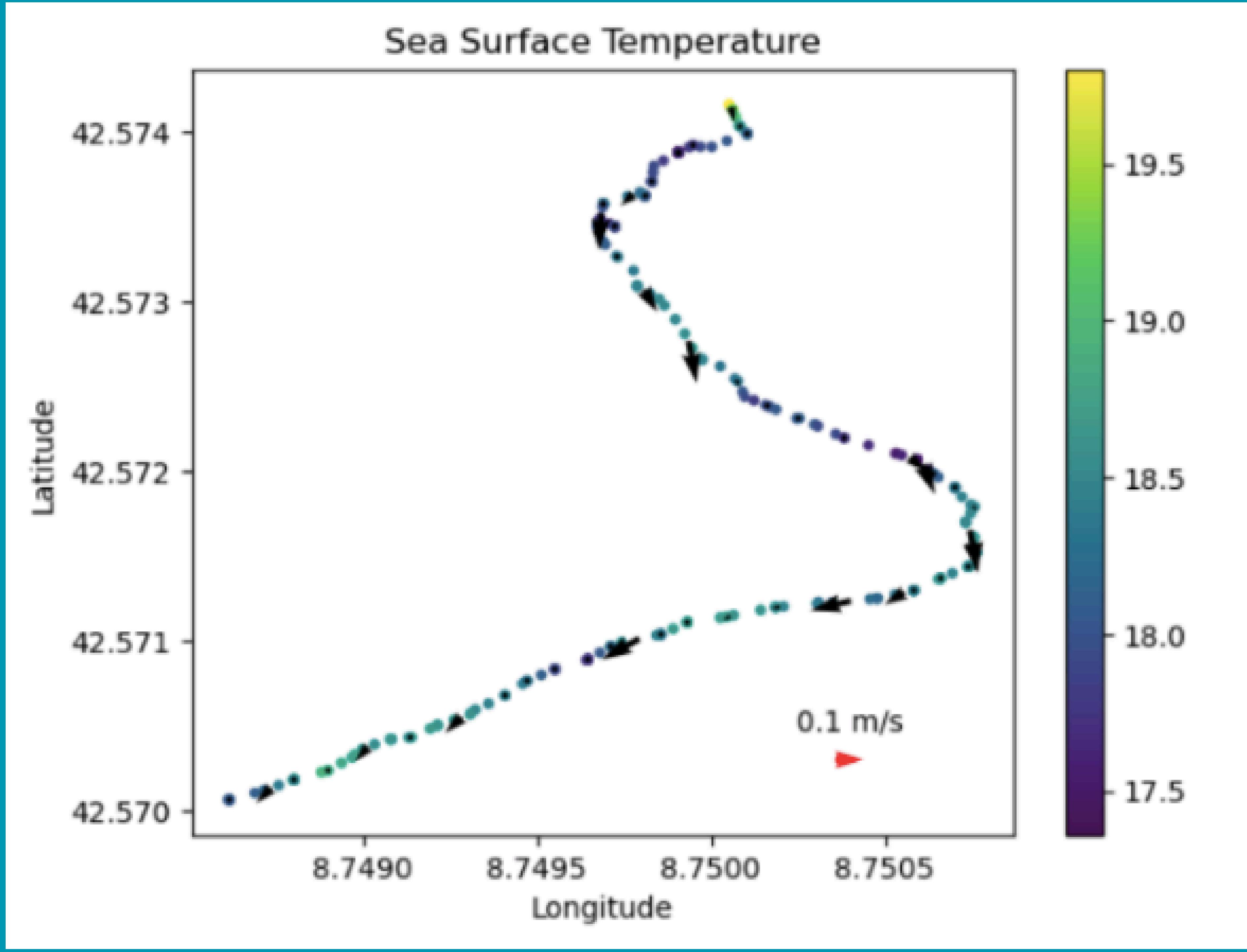


Figure 11: Surface temperatures as a function of longitude and latitude. Surface current strength is shown by black arrows.

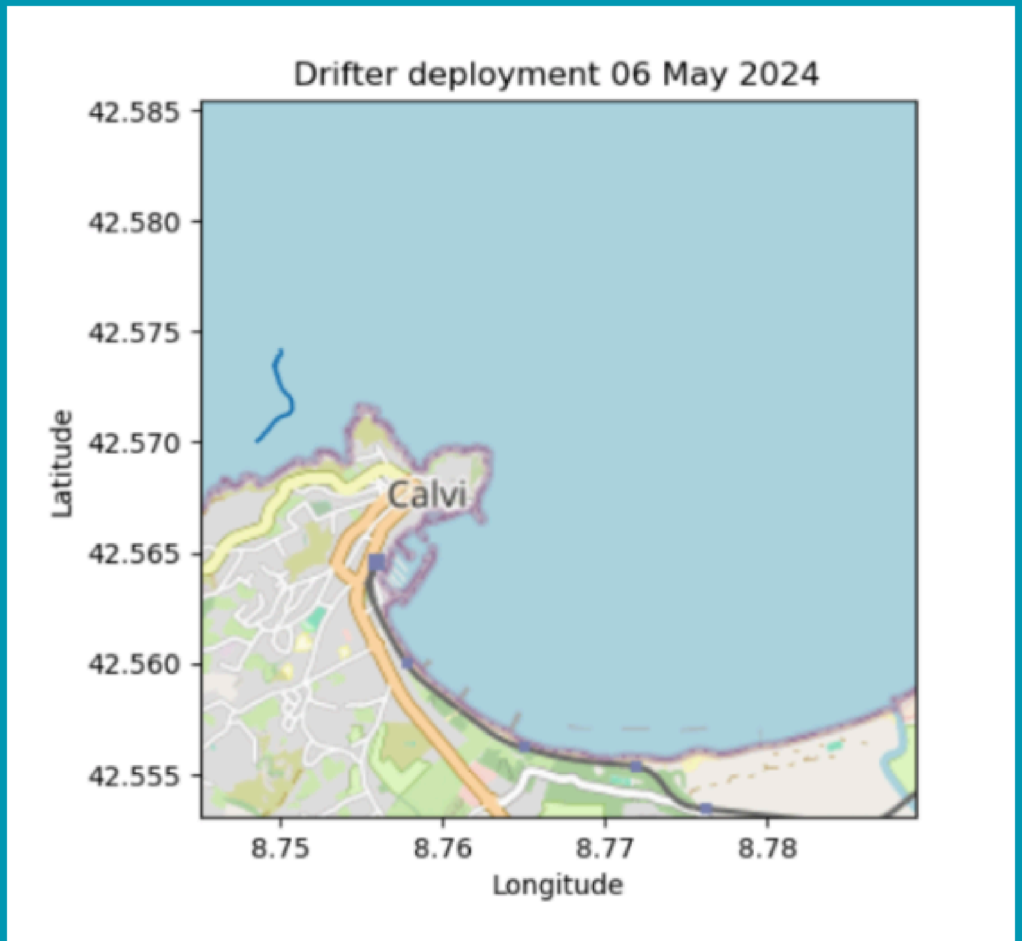


Figure 10: Map showing the movements of Drifter 02 in Calvi Bay on May 6, 2024.

- **Warmer water near the coast: +1°C**
- **Consistent currents**
- **Currents weaken near the coast**
- **Current speed: ±0.2 m/s**

Analysis of Results: Drifter, May 2–6

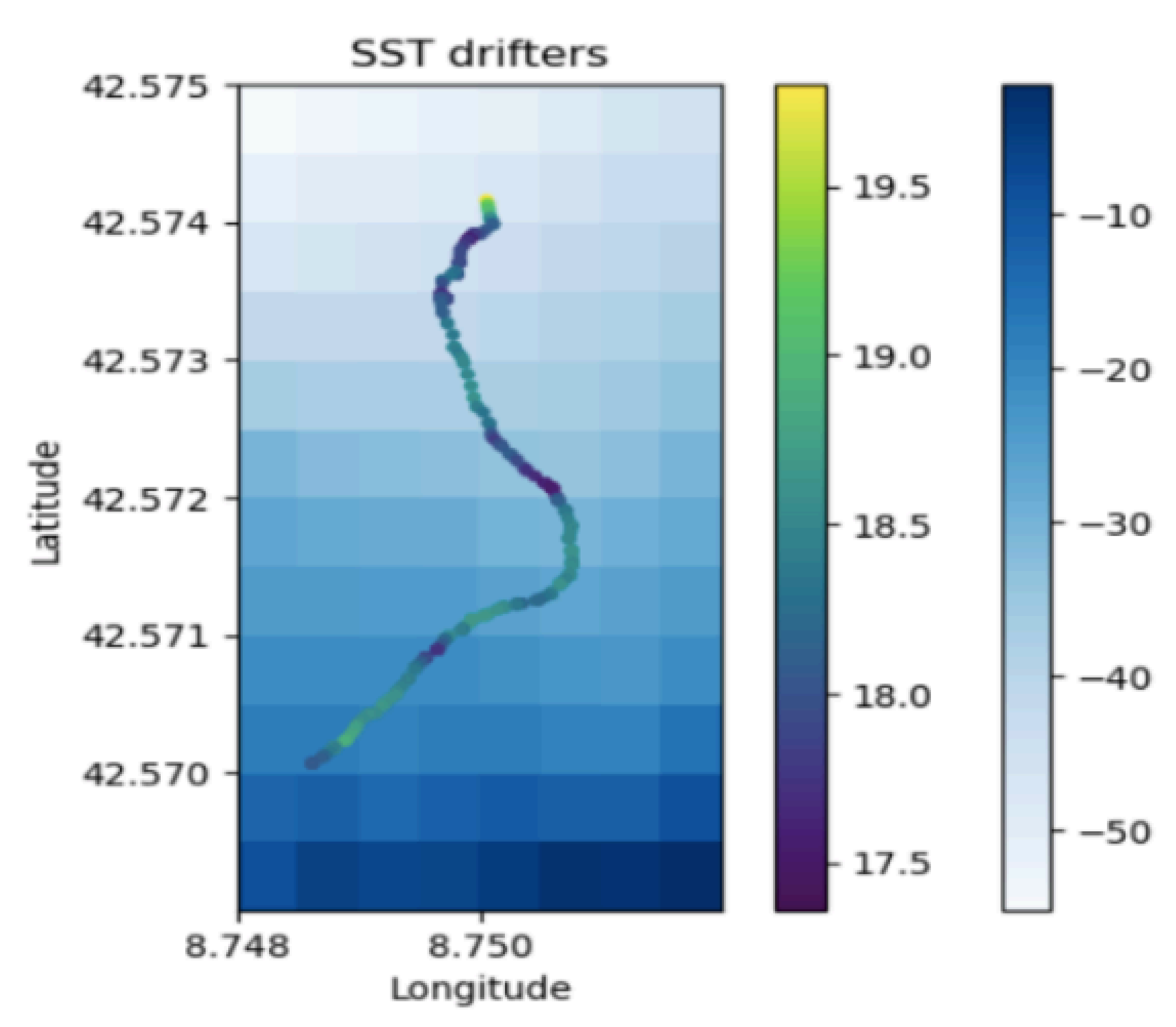


Figure 12: Surface temperatures as a function of GPS location. Bathymetry is shown using the blue scale.

- **Warmer water near the coast: +1°C**
  - **Less obvious warming of the water**
  - **Greater temperature variations during the drift**

- **The relationship between water temperature and bathymetry is much less obvious**

Analysis of Results: Drifter, May 2–8

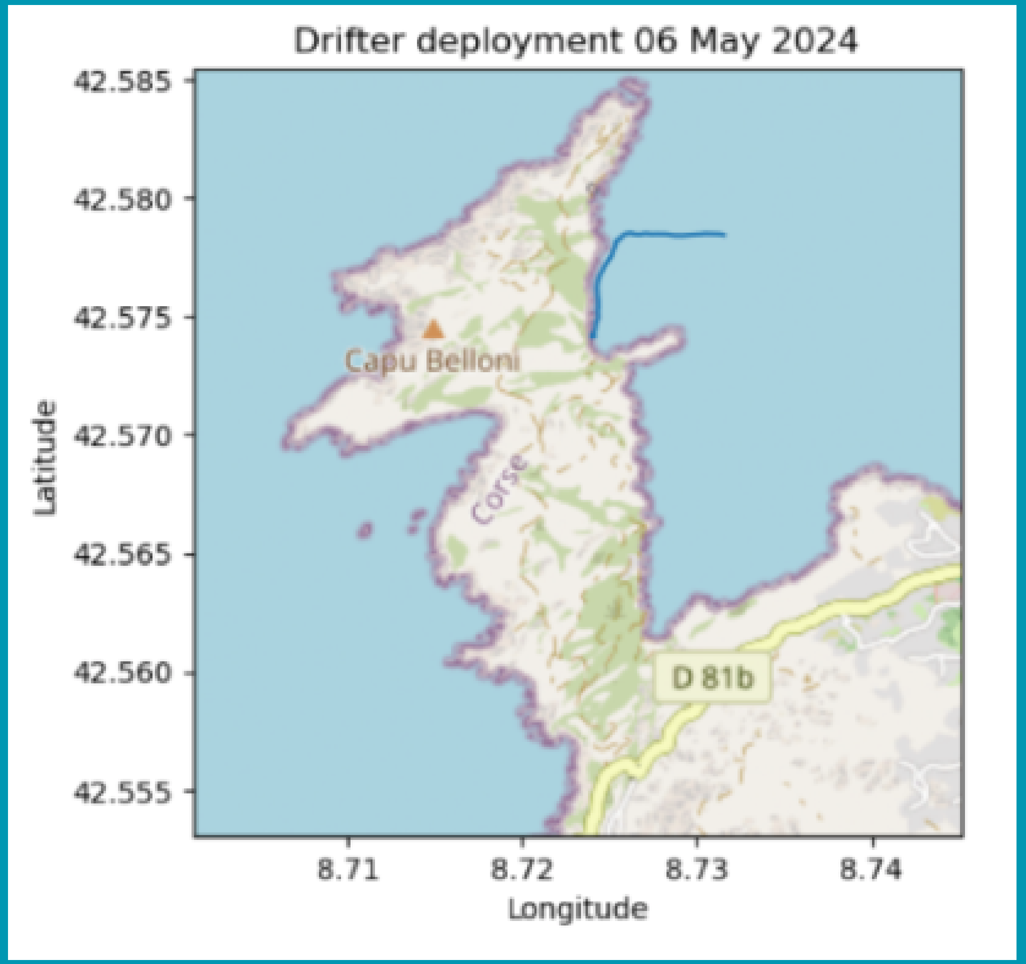


Figure 13: Map showing the movements of Drifter 02 in Calvi Bay on May 8, 2024.

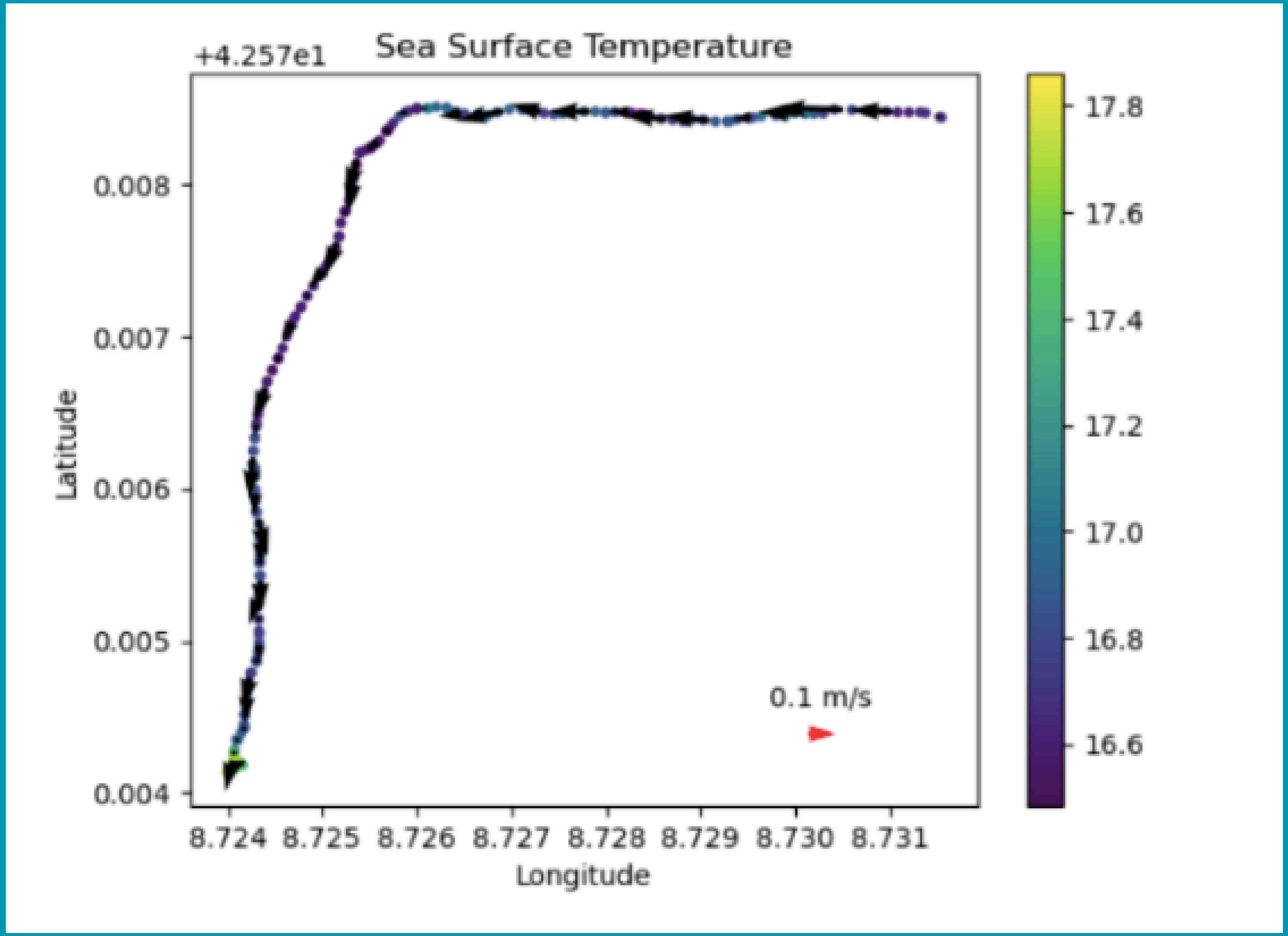


Figure 14: Surface temperatures as a function of longitude and latitude. Surface current strength is shown by black arrows.

- **Temperature change: +1.4°C**
- **Low current speeds: 0.1 m/s**
- **Currents consistent with the drifter's direction**

Analysis of Results: Drifter, May 2–8

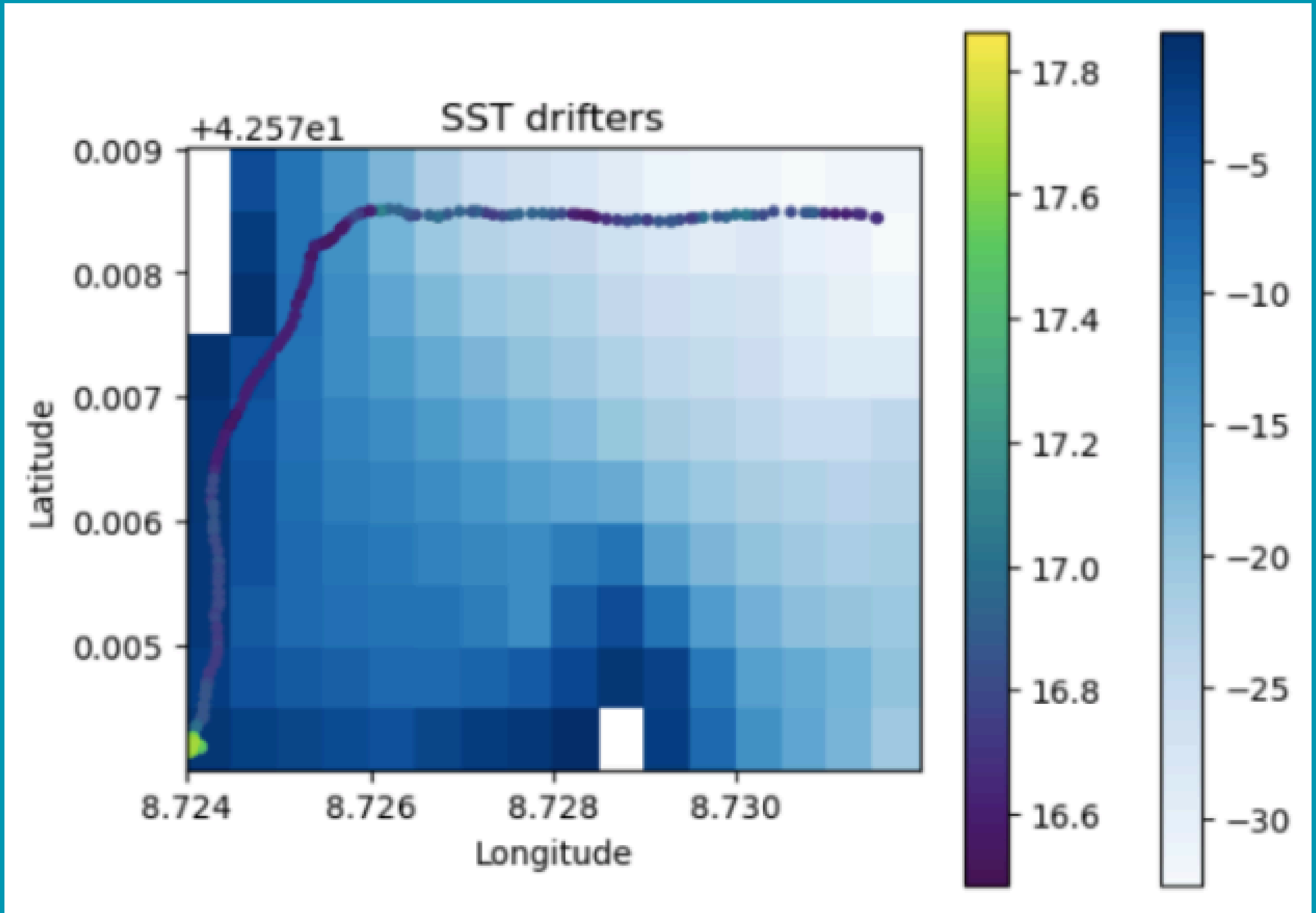
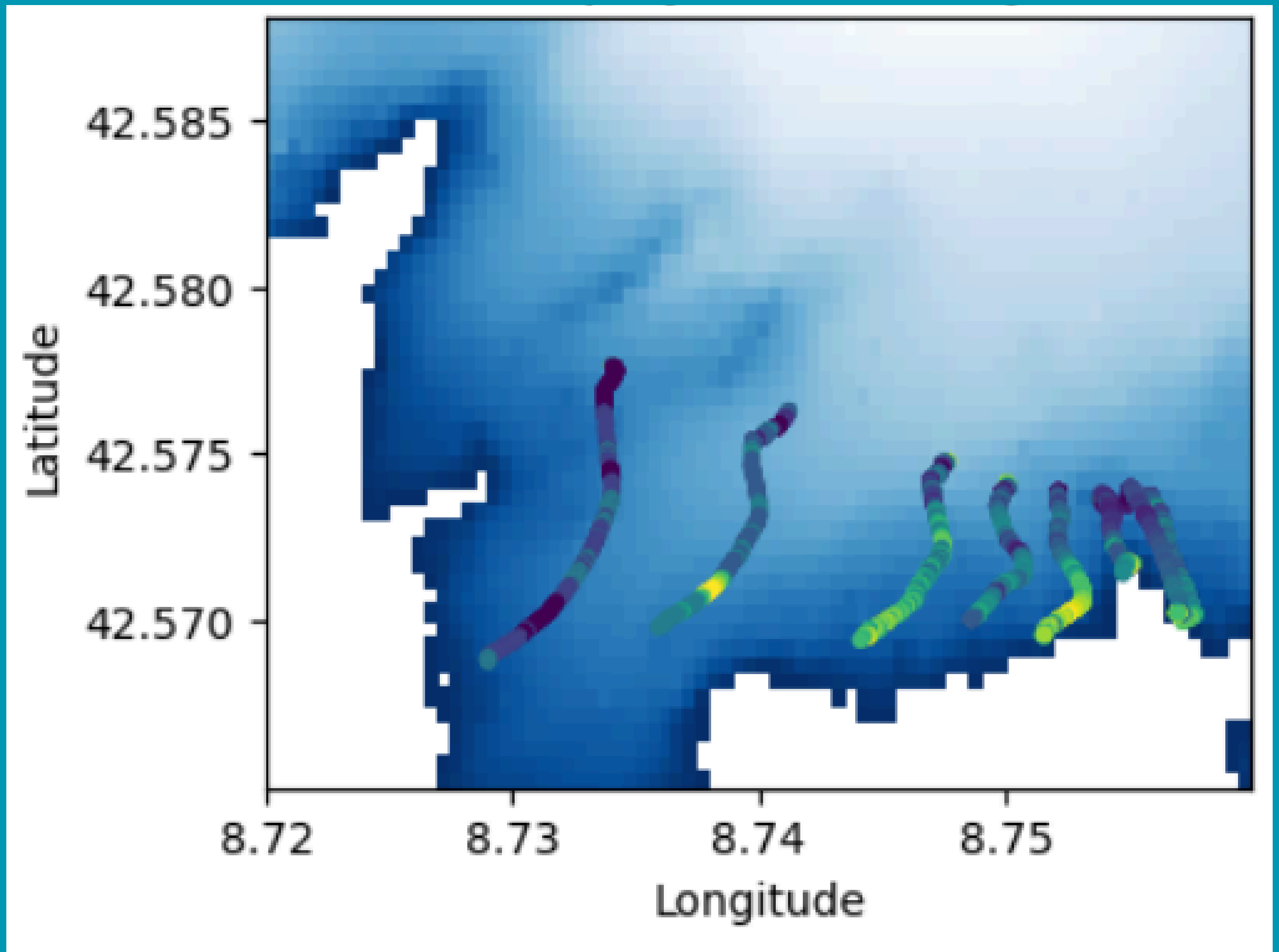


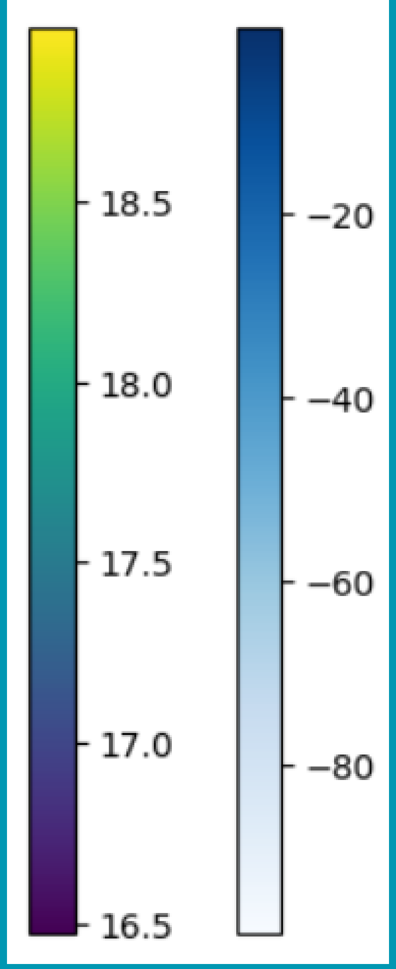
Figure 15: Températures de surface en fonction de la localisation GPS. Bathymétrie représentée avec l'échelle bleue.

- **Temperature change: +1.4°C**
- **Cooling near the coast**
  - **Current/upwelling of cold water ?**
  - **Link between bathymetry and temperature not clear**
- **Warming toward the end of the route**
  - **drifting among the rocks**

Comparison with other drifters - Temperature

May 6





May 8

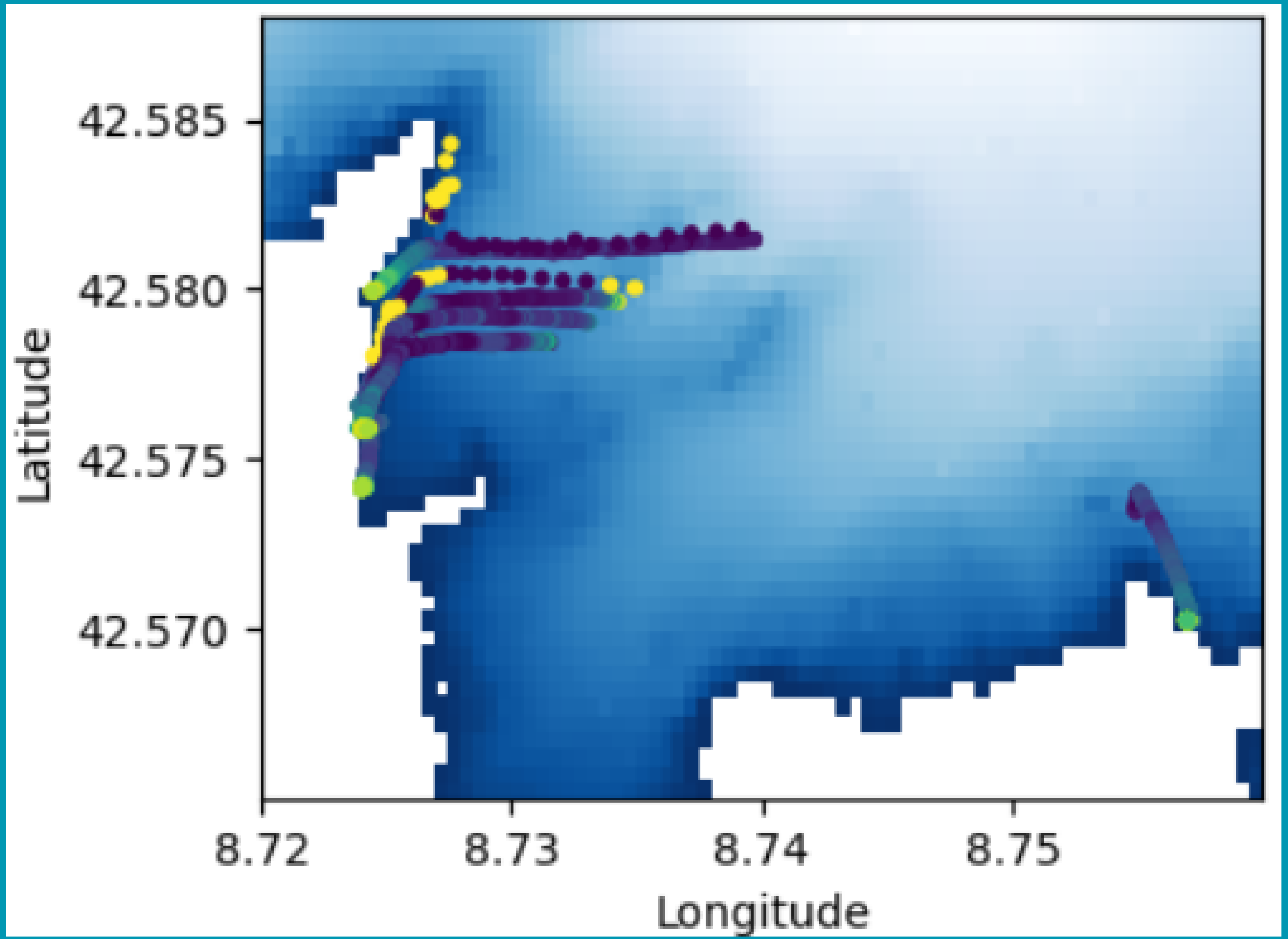


Figure 16: Overview of the drifters' trajectories, temperature, and bathymetry

Figure 17: Overview of the drifters' trajectories, temperature, and bathymetry

- Consistency among the drifters
- Variation of +/- 1.5°C

Comparison with other drifters - Movements

May 6

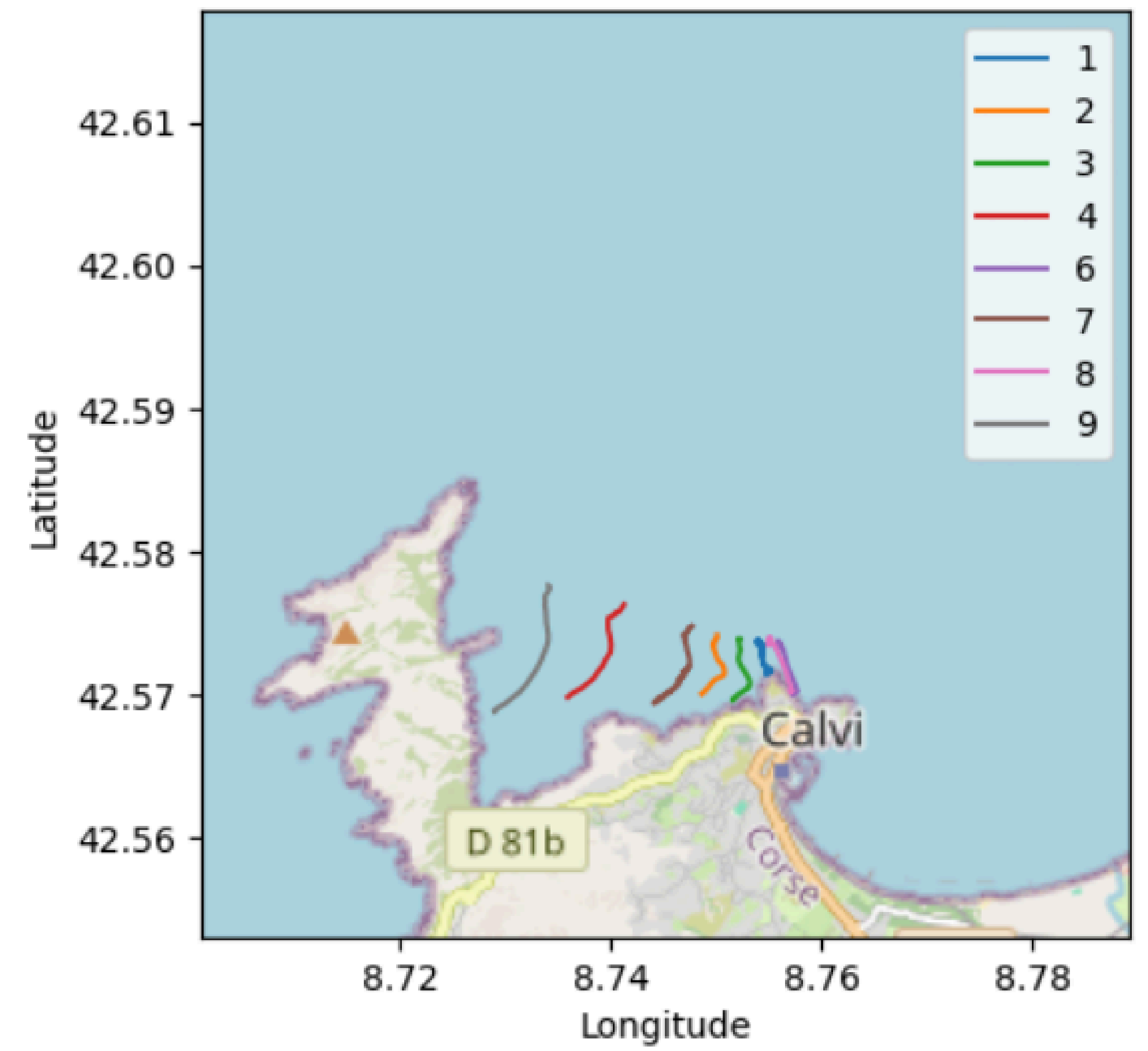


Figure 18: Overview of the drifters' trajectory (May 6)

May 8

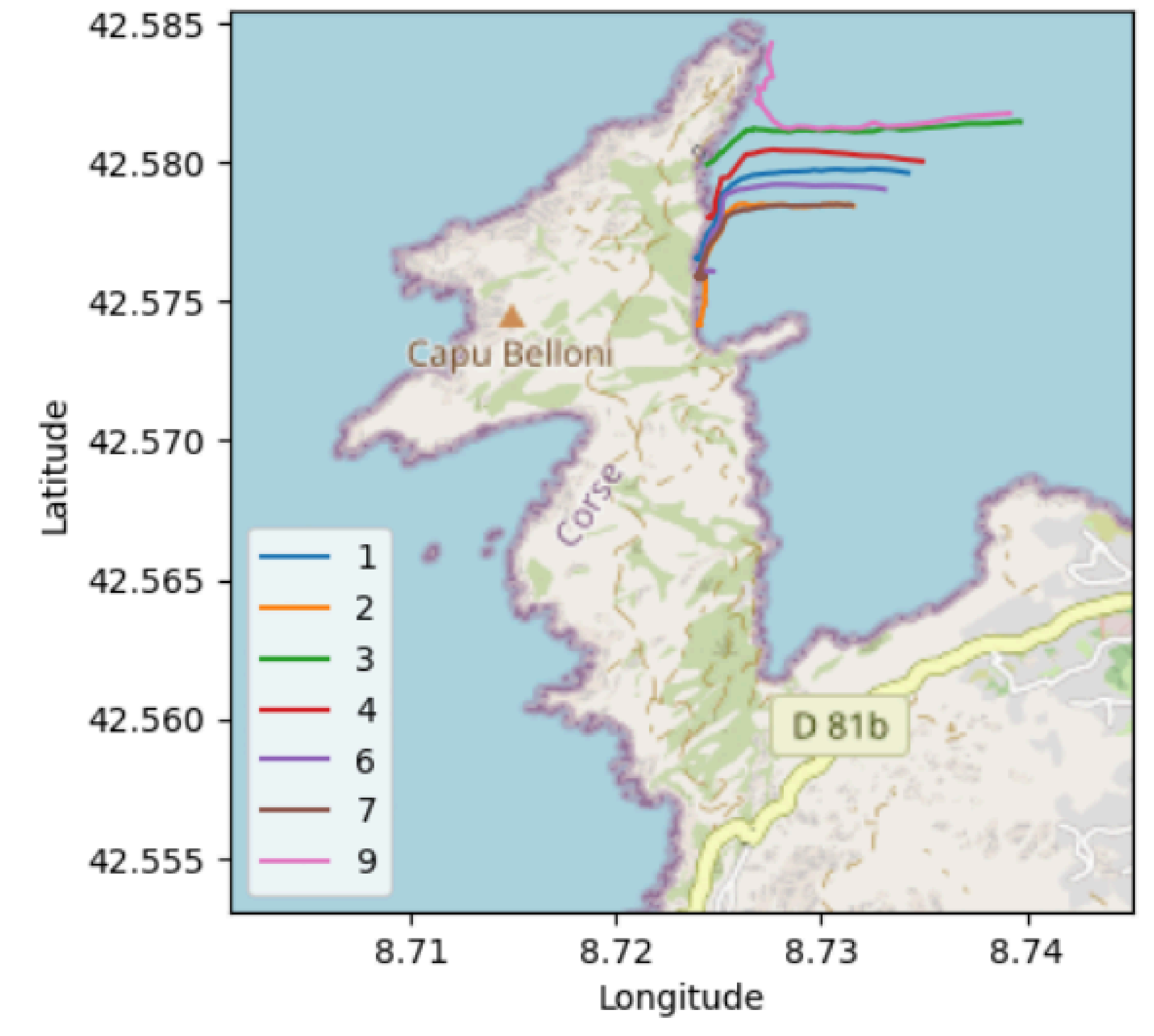


Figure 19: Overview of the drifters' trajectory (May 8)

- Follow the same trajectory depending on the current
- May 8: Drifter 9 drifting northward

Suggestion for improving the drifter

Mast and Sails:

- Requires a larger space to build them
- Fun activity
- More durable materials (iron bars)

Strategy:

- Develops critical thinking and scientific reasoning
- Applies the theory learned in Q1
- Adjusts the drop (wind, etc.)

Coding:

- Expectations that are far beyond our capabilities
- A feeling of being a bit lost

To do:

- A data reading session
- Salinity sensor
- Determining wind direction


Thank you !
Any questions ?

Division of labor

- **Raspberry Pi Programming:** All together
- **Drifter Construction:** All together
- **Deployment Strategy:** Violette/Ylenia/Anne
- **Poster:** Anne/Océane
- **Graphs/Figures:** Violette
- **Description/Interpretation of Results:** Anne/Océane/Ylenia
- **Suggestions for Improvement/Conclusion:** All together